

GH0ST_B0Y Service VM-LLM AR Augmentation Project Summary

This document summarizes your vision for a custom augmented intelligence system integrated with AR glasses, based on our conversation. It includes key hardware research, sensor augmentations for human senses (e.g., heat/IR for pre-touch safety, motion for equilibrium, heart rate/temperature for health monitoring, holographic displays for visual overlays, and voice activation), privacy-focused encryption, and open-source communities to recruit AR devs. The goal is to simplify and safer human life through a secure, personal "neural link" to your VM-LLM engine without invasive devices. Designed for your dev team to prototype and build upon, emphasizing open-source modding for the GH0ST_B0Y Service (a virtual machine-based LLM with human-centric services like sense augmentation, real-time safety alerts, and personalized AI interactions).

Your Vision

Researched Hardware Options

Based on your preferences for open-source/moddable AR glasses with heat, IR, motion sensors, cameras, sound, and 3D/depth perception:

ThirdEye X2 MR Glasses (\$2,450 via thirdyegen.com or ozrobotics.com):

Specs: Built-in thermal/IR sensors (expandable via USB-C, e.g., FLIR modules), 13MP camera, 9DoF IMU motion sensors, mics/speakers for voice, 6DoF SLAM for 3D depth mapping.

Why Fits: Industrial-grade for safety apps; SDK for custom interfaces/privacy. Mod for your touch simulation and heart rate add-ons.

UTRACK AI Outdoor Glasses (\$2,299 via [Amazon](https://amazon.com) or utechwear.com):

Specs: 384x288 thermal/IR sensor (up to 1,500m detection), 48MP camera, IMU motion tracking, sonar/sound integration, AI for multi-layer perception (thermal + visible + depth).

Why Fits: Outdoor/safety focus; local AI for privacy; moddable for equilibrium alerts and voltage touch.

Brilliant Labs Halo (\$299 via brilliant.xyz; open-source):

Specs: Monocular holographic display, camera/mic for voice/AI, expandable for IR/motion/heart rate sensors via community mods.

Why Fits: Fully open-source (GitHub schematics/firmware); ideal starting point for custom VM-LLM integration without proprietary locks.

Sensor & Augmentation Details

Heat/IR/Temperature: Use MLX90614 or FLIR Lepton for object scanning; AR overlays warn of temperatures (e.g., "Cold: 32°F") to prevent accidents. **Motion/Equilibrium:** 6-axis IMU (BMI088) for balance detection; alerts via haptic/voice if imbalance. **Touch Simulation:** Capacitive voltage sensor (e.g., AT42QT1010) for "static feel" based on IR data. **Other:** Heart rate via earlobe clip; all synced to GH0ST_B0Y VM-LLM for AI processing (e.g., "High heart rate—relax?"). **Software Stack:** Local LLMs (Llama/Mistral), LangChain for agents, Bluetooth for sensor hubs (ESP32); E2EE for user accounts.

Open-Source Communities for Dev Team Building

To recruit AR-specific devs for GH0ST_B0Y (focus on VM-LLM integration with human sense augmentation), engage these communities—they're active in 2026 for open AR hardware/software:

Team Open Smart Glasses / Mentra Community: GitHub (Mentra-Community/OpenSourceSmartGlasses, MentraOS); Discord (<https://discord.gg/5ukNvkEAqT> or <https://mentra.glass/discord>). Focus: Open-source smart glasses SDKs, app stores, hardware designs. github.com +3 Based Hardware / OpenGlass: GitHub (BasedHardware/OpenGlass); Discord (<https://discord.com/invite/ZutWMTJnwA>). Low-cost DIY smart glasses mods. github.com Brilliant Labs: GitHub (brilliantlabsAR); Discord (<https://discord.gg/7w3DFxek4p>). AR/AI glasses hacking and projects. github.com +1 General XR/AR Forums: Reddit r/augmentedreality; Smart Glasses Community (<https://smartglasses.community/>); XR Creators Discord (<https://discord.com/invite/27v2E92sjw>). github.com +1 Post job listings or collaborate on GitHub forks for sensor integrations.

Next Steps for Dev Team

Prototype with Halo or ThirdEye as base; add sensors via ESP32. Integrate GHOST_BOY VM-LLM: Use Python (Ollama, TensorFlow Lite) for edge AI. Recruit: Post in above Discords/GitHub issues seeking AR devs experienced in Unity/Android Studio for holographic UIs and sensor fusion. Test: Focus on safety (e.g., IR accuracy $\pm 2^{\circ}\text{C}$) and privacy (biometric auth). Contact me for more research or script tweaks.